Was magnitude estimation necessary?

A secondary-data reassessment of Bard, Robertson, and Sorace (1996)

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Abstract

Was magnitude estimation necessary to establish acceptability judgments as serious linguistic measurements? I separate Bard, Robertson, and Sorace's broader case for graded acceptability from the narrower claim that magnitude estimation improves on simpler formal tasks. Using public method-comparison data from Sprouse, Schütze, and Almeida (2013) and Sprouse and Almeida (2017), I compare magnitude estimation with Likert, forced-choice, and yes/no judgments. The primary analysis follows a charter fixed before outcome analysis. I find a strong common item-level signal and the bounded Likert curvature that motivates magnitude estimation, but no systematic practical advantage unique to magnitude estimation in endpoint, prediction, or decision diagnostics. A post-outcome multiverse of 162 specifications confirms the result. Four 2017 rank-based forced-choice models show small incremental predictive gains, but these don't generalize to raw-scale models, yes/no judgments, or the 2013 data. The results preserve Bard et al.'s broader contribution (i.e., the formal measurement of graded acceptability) while weakening the case that magnitude estimation has a general practical advantage for the contrasts examined.

Keywords: magnitude estimation; acceptability judgments; grammaticality judgments; secondary data; experimental syntax

AI use. The large language models OpenAI GPT-5 and gpt-5.6-sol (Codex); Anthropic Claude Opus served as drafting and editing aids throughout the preparation of this paper. I am responsible for all theoretical claims, arguments, errors, and interpretive choices.

1 The problem

Bard, Robertson, and Sorace made a methodological claim that was larger than a single elicitation format. Acceptability judgments, they argued, could be treated as graded measurements rather than as informal binary verdicts (Bard et al., 1996). That move helped make experimental syntax a measurement enterprise: judgments could be collected from naive participants, scaled, compared across groups, and subjected to ordinary statistical tests.

Whether magnitude estimation was necessary for that advance is a harder question. Magnitude estimation promised a way around the scale restrictions of traditional judgment tasks, giving participants an open response scale and preserving fine distinctions. But later acceptability work didn't settle into one privileged elicitation method. Likert ratings, forced choice, yes/no tasks, and Thurstonian models all became part of the measurement toolkit (Langsford et al., 2018; Sprouse & Almeida, 2017; Sprouse et al., 2013).

This paper separates two claims that are often allowed to travel together. One claim is strong and still plausible: acceptability is graded, measurable, and methodologically serious. The other is weaker: magnitude estimation has a practical measurement advantage over simpler formal tasks. The first claim can survive even if the second doesn't.

This empirical question isn't new. The methodology literature has already compared elicitation formats and largely deflated magnitude estimation's special status. Schütze (2016) reports that most or all of its advertised advantages over Likert tasks have since been refuted, pointing to Weskott and Fanselow (2011) and Sprouse et al. (2013); later work weighed the formats on convergence (Sprouse et al., 2013), statistical power (Sprouse & Almeida, 2017), and reliability and response style (Langsford et al., 2018).

This paper instead contributes a clearer routing of the evidence. Existing method-comparison work already weakens magnitude estimation's special status; the reassessment asks what the surviving formal-judgment signal licenses us to project. In the approved reanalysis of the 2013 and 2017 comparison data and a post-outcome robustness multiverse, formal judgment formats preserve stable item-level contrast ordering across the tested contrasts, while magnitude estimation adds no robust separate practical ordering there.

Seen this way, the reassessment doesn't reject Bard et al.'s programme. It tests what should be credited to magnitude estimation itself, and what should instead be credited to the broader formalization of acceptability judgment methodology.

2 The Bard target

Reassessing Bard et al. means first stating, from the article rather than its reputation, what they actually claimed. The abstract makes four claims for magnitude estimation: that it can "solve the measurement scale problems which plague conventional techniques", "provide data which make fine distinctions robustly enough to yield statistically significant results of linguistic interest", be "usable in a consistent way by linguistically naive speaker-hearers", and "allow replication across groups of subjects" (Bard et al., 1996, p. 32).

The core of the case is an argument about scale type, not about reliability. The trouble with conventional annotation, they argue, is "the use of the wrong kind of measurement scale" (Bard et al., 1996, p. 32): the asterisk system, like any fixed-point scale, "predetermines the number of distinctions subjects may use" and so censors finer ones (Bard et al., 1996, p. 35). Magnitude estimation is the proposed remedy because it "does not restrict the number of values", giving subjects "complete freedom about which of the infinite set of numbers to use" and yielding interval and ratio data (Bard et al., 1996, p. 41).

This sharpens Bard et al.'s claim and its boundary. They do make a method-superiority claim, but a specific one: an open, ratio-yielding response recovers gradient distinctions that fixed-point scales lose by construction. The claim concerns resolution and scale-boundary censoring, and it can be tested. The broader reading sometimes met in reception, that magnitude estimation is the privileged formal method in general, goes beyond the article; Bard et al. compare magnitude estimation with informal annotation and fixed-point scales, not with the later inventory of Likert, forced-choice, yes/no, and Thurstonian tasks.

Bard et al. are also more tentative than the reception. They grant that the advantages "are garnered at considerable empirical cost", and they leave open which psychological model underlies acceptability, since "the data currently do not decide between them" (Bard et al., 1996, p. 65). The fair target for reassessment is the resolution claim, not a gold-standard claim the authors didn't make.

The distinction matters for design. If the live claim is that an open ratio scale recovers distinctions fixed scales censor, then the later method comparisons have to test resolution and scale-boundary effects alongside convergence and reliability. Because Bard participant-level rows aren't available to this project, the paper can't replicate the original experiment directly; the reassessment is a conceptual replication and secondary-data test of whether the resolution claim retains independent evidential force.

3 Secondary-data design

The secondary-data design treats the project as a measurement reassessment of Bard, Robertson, and Sorace (Bard et al., 1996). It separates four kinds of source from the observation channels introduced in Section 6: sources are routed to estimands, while channels bear on grammatical inferences. Published summaries from Bard et al. can source the historical claim, but can't support a direct replication without their participant-level data.

Later judgment studies can support secondary analysis when they compare elicitation methods on shared or comparable items and provide the rows needed for the intended estimand. Computational benchmarks can show the afterlife of acceptability labels, but they don't bear on magnitude estimation versus Likert ratings. Construct-level discussion can separate acceptability from grammaticality, naturalness, standardness, and correction.

Source	Evidence level	Use in this paper	Unsupported claim
Bard et al. 1996	Published article	Historical target and scale-resolution claim	Direct replication without participant rows
Sprouse, Schütze, and Almeida 2013; Sprouse and Almeida 2017	Public row-level comparison data	Constrained secondary method comparison within approved reuse boundary	Raw-data redistribution or drift beyond the approved methodological scope
Langsford et al. 2018	Published article and supplement	Reliability and response-style synthesis	New participant-level reliability model without raw rows
CoLA, BLiMP, SAD	Benchmark labels and documentation	Afterlife of formal acceptability measurement	Evidence about ME versus other human judgment methods

Table 1: Evidential routing before analysis. ME = magnitude estimation; SAD = Syntactic Acceptability Dataset.

The approved reuse boundary resolves the secondary-data side of the genre gate. Public row-level data from the two comparison studies support a constrained secondary analysis of method performance. The original studies had different aims: Sprouse et al. (2013) estimated convergence between informal and formal judgments, while Sprouse and Almeida (2017) studied task sensitivity and statistical power. The present resolution analysis is a secondary use of their rows, not a claim about what those studies were designed to test.

The verified public Langsford materials available to this project lack raw participant responses. This limits Langsford et al. to the article-summary level; the analysis doesn't attempt a participant-level response-style or test-retest model across the literature.

The comparison files also require representation choices before modelling. For the 2013 data, the primary forced-choice comparison uses the pair-level sign-test file because forced choice is a pairwise task and that file records judgments against expected patterns. The logistic file preserves item-level binary rows and changes the unit of analysis, so it serves as a named sensitivity specification.

For the 2017 data, yes/no rows with item IDs `B`, `G`, and `M` are excluded from item-level comparisons. They're absent from the materials workbook, occur in repeated early order positions for every subject, and don't align with the magnitude-estimation, Likert, or forced-choice item universe. This exclusion is a structural inclusion rule, not an outcome-based trimming decision.

Each source is assigned to an estimand it can answer. The primary estimand is the item-level acceptability signal recoverable across methods. Secondary estimands are method-specific noise or bias and decision agreement for theoretically relevant contrasts.

Participant response style and method-by-item instability are admissible only when the dataset contains the participant-level rows and repeated structure needed to estimate them. This ordering keeps the analysis from selecting whichever scale looks best after the fact, the kind of forking path Gelman and Loken warn about (Gelman & Loken, 2013, 2014).

The substantive comparison uses a measurement null, but a careful one. Magnitude estimation, Likert ratings, forced choice, and Thurstonian choice are modelled as readings of a common item-level acceptability signal through method-specific response functions, not as the same signal plus method noise: the bounded methods can compress the signal at the extremes, while the open ratio scale need not. The single-dimension assumption is checked before any method-specific term is read. With richer row-level evidence, that check would compare a one-factor model against correlated-two-factor and bifactor alternatives and inspect method residuals for item-level structure. In the present reanalysis, the implemented counterpart is deliberately weaker: a first-pass PCA/parallel-analysis check over aggregate item and contrast matrices.

This ordering matters because the advantage Bard et al. actually claimed is about resolution, recovering distinctions that fixed scales censor (Bard et al., 1996, p. 35). A plain common-signal-plus-noise null would book that resolution as magnitude-estimation noise and let the method only lose; the response functions and the dimensionality check let a resolution advantage register instead. If one factor suffices, magnitude estimation still has to earn its term; if it doesn't, the analysis turns to where the methods diverge. Scrambled labels, method-label permutations, and simulations from the response-function model would be pipeline controls rather than substantive comparisons; the paper doesn't use them as evidence about ME.

After the chartered analyses were complete, a bounded multiverse tested whether their conclusion depended on defensible representation choices. Its specification universe was frozen before that analysis was run, but after the primary outcomes were known, so it's a post-outcome robustness analysis rather than a prespecified test. Three families vary ME and Likert scoring, endpoint definitions and spread measures, raw- and rank-scale prediction, and independent forced-choice or yes/no adjudication. The resulting 162 specifications test scale-boundary spread, incremental out-of-sample information, and actual contrast-sign disagreements without treating unexplained ME variance as an advantage by itself. The complete decision rules and specification list are versioned in `notes/sprouse-resolution-multiverse-plan.md`.

4 Scale and reliability

The scale question is practical measurement advantage over the response-function measurement null set out in the secondary-data design. Bard et al.'s case for magnitude estimation included scale resolution, fine-grained distinctions, naive-participant use, and replication across groups (Bard et al., 1996, p. 32). The safer target is that source-grounded claim, together with the stronger reading of magnitude estimation that took hold in parts of the later methodology literature.

Later method-comparison data give the paper a sharper question: does magnitude estimation recover item differences, participant agreement, or theoretically relevant decisions that simpler tasks lose, and in particular does it recover graded distinctions that bounded scales compress at the extremes?

Sprouse et al. (2013) compare formal judgment tasks with informal published judgments, giving evidence about convergence between magnitude estimation, Likert ratings, and forced choice. The public rows from that study can support a bounded reanalysis of item-level agreement and decision consequences within the approved reuse boundary. Langsford et al. (2018) add a reliability and response-bias target by comparing Likert, magnitude estimation, forced choice, and Thurstonian models. Without their raw rows, Langsford can support the synthesis of published reliability findings but not a new participant-level response-style or test-retest model.

The reanalysis inherits the unit-of-analysis charter fixed in the secondary-data design rather than reopening it. The same primary and sensitivity representations carry over, so the claims stay traceable to fixed units of analysis instead of to whichever representation later proves most convenient.

The primary comparison weighs methods on resolution, prediction, decision consequences, and convergence, prioritizing consequences for the linguistic contrast. Resolution is the criterion Bard's claim lives on: whether magnitude estimation recovers graded item differences that bounded scales compress at ceiling or floor, or a measurement dimension that survives the dimensionality check. Reliability is a distinct property, not a proxy for it; Langsford et al. (2018) report magnitude estimation as less reliable, which doesn't by itself settle resolution.

For this paper, a practical magnitude-estimation advantage would have to change an inferential decision. A summary-statistic improvement alone wouldn't count. An advantage would count if magnitude estimation recovered a good-vs-bad contrast that bounded methods missed, showed its largest extra spread exactly where bounded scales compressed the response, carried a stable method-specific dimension across scoring choices, or reduced sign-error or exaggeration risk for a named contrast under a design analysis. A larger correlation, by itself, wouldn't settle the question.

The reanalysis of both comparison datasets gives a narrow answer. Magnitude estimation and the bounded methods share a strong aggregate item-level signal. Raw Likert means also show the expected bounded response-function curvature relative to magnitude estimation. That curvature is the pattern Bard's scale argument makes visible, but the endpoint and pair-level diagnostics don't show a systematic practical distinction preserved only by magnitude estimation.

Figure 1 makes the common signal and bounded curvature visible. The linear fit tracks the centre of each cloud but runs past the Likert scale's upper boundary, while the bounded fit turns towards that boundary. That pattern is why the analysis allows method-specific response functions before asking whether magnitude estimation preserves a further practical distinction.

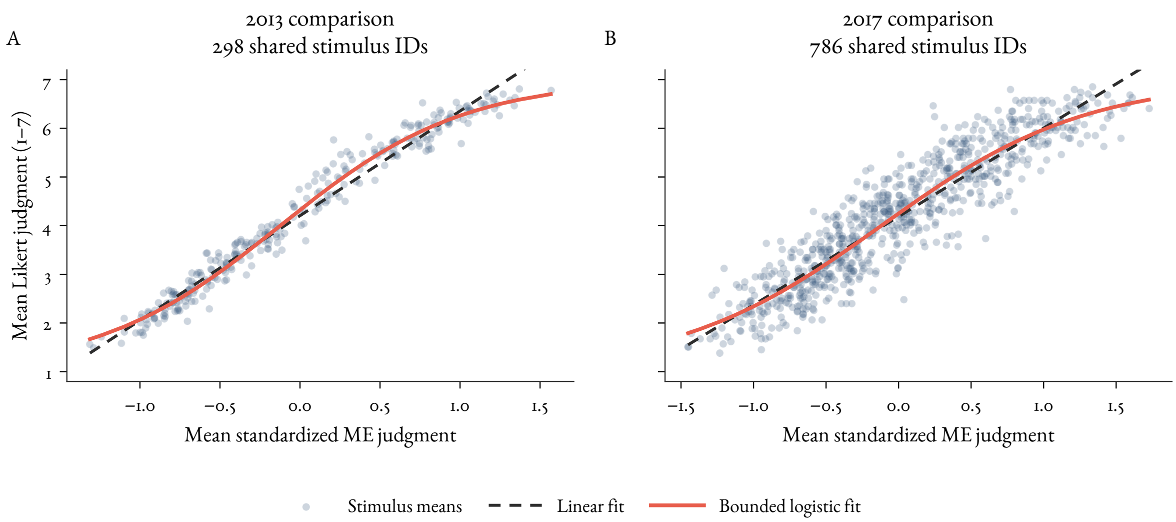

Figure 1: Aggregate response functions in the public comparison data from Sprouse, Schütze, and Almeida (2013) (A) and Sprouse and Almeida (2017) (B). Each point averages participant judgments for one stimulus identifier shared by the magnitude-estimation and Likert files. The 2013 files call these identifiers "conditions" (298 points); the 2017 files call them "items" (786 points). The two labels reproduce the source-file nomenclature. The dashed line is a linear fit and the solid line is a 1–7 bounded logistic fit. Magnitude estimation is an unbounded diagnostic axis here, not the assumed true latent acceptability scale.

The dimensionality result is deliberately modest: a first-pass aggregate check of the item and contrast matrices, used to ask whether a clear multidimensional rescue for magnitude estimation appears in the available aggregate summaries. It doesn't replace a full participant-level factor model, which would require a broader row-level evidence base than this paper has.

The rows in Table 2 come from the approved pipeline documented in `analysis/README.md`: item-signal, dimensionality, response-function, pair-robustness, and aggregate sensitivity scripts. The reported n values are aggregate condition, item, or pair units, not participant-level rows. A 95% percentile bootstrap over those same aggregate units gives intervals that leave the interpretation unchanged: ME–Likert correlations remain high for 2013 conditions [.983, .989], 2013 pairs [.950, .974], 2017 items [.915, .934], and 2017 pairs [.917, .965], and pair-level bounded-predictor R^2 remains high for 2013 [.900, .946] and 2017 [.862, .941].

Diagnostic	2013 study	2017 study
ME–Likert convergence	Conditions $n = 298$, $r = .986$; pairs $n = 150$, $r = .963$	Items $n = 786$, $r = .925$; pairs $n = 50$, $r = .944$
First-pass dimensionality check	Pairs $n = 150$: first component 89.4%; second eigenvalue below parallel-analysis threshold	Items $n = 786$, pairs $n = 50$: first components 80.6% and 85.9%; second eigenvalues below thresholds
Bounded response function	Conditions $n = 298$: bounded logistic improves over linear; endpoint ME SDs .142 and .175 versus middle .313	Items $n = 786$: bounded logistic improves over linear; endpoint ME SDs .249 and .321 versus middle .349
Pair robustness	Pairs $n = 150$: bounded predictors $R^2 = .926$; no ME top-quartile pair falls in the bounded bottom half	Pairs $n = 50$: bounded predictors $R^2 = .901$; no ME top-quartile pair falls in the bounded bottom half

Table 2: Primary diagnostics from the comparison data accompanying Sprouse, Schütze, and Almeida (2013) and Sprouse and Almeida (2017) for Bard's resolution claim. ME = magnitude estimation. The dimensionality row is an aggregate first-pass check, and the response-function rows use magnitude estimation as an unbounded diagnostic axis rather than the true latent acceptability scale.

Table 2 separates three facts. First, the aggregate matrices provide no obvious multidimensional rescue for magnitude estimation. Second, the Likert scale behaves like a bounded response function, so a fair null should allow compression. Third, the available contrasts don't show the further pattern that would make magnitude estimation practically superior: endpoint regions aren't where magnitude estimation spreads out most, and good-vs-bad pairs ranked highly by magnitude estimation are also ranked highly by bounded methods.

The explicitly post-outcome multiverse asks whether that interpretation survives defensible scoring and modelling choices. It varies three ME scores, three Likert scores, endpoint definitions and spread statistics, raw- and rank-scale prediction, and forced-choice or yes/no adjudication. None of its endpoint, incremental-prediction, or decision-adjudication families meets the frozen family-level criterion in either dataset.

The disaggregated result contains one useful qualification. Both endpoint ratios exceed one in 3 of 27 specifications for 2013 and 6 of 27 for 2017; only one 2017 specification also clears the practical and permutation criteria. ME adds at least .02 cross-validated R^2 beyond Likert in 0 of 36 prediction specifications for 2013 and 4 of 36 for 2017. Those four are rank-scale models of forced choice ($\Delta R^2 = .025$–$.049$); the increment doesn't generalize to 2017 yes/no judgments or to raw-scale mappings. ME and Likert disagree on the contrast sign for only 3 of 150 pairs in 2013 and 1 of 50 in 2017, too few to clear the decision criterion. The localized 2017 result makes a blanket no-information claim too strong for those specifications, but it doesn't establish a systematic practical advantage.

Figure 2 shows why the qualification matters without changing the family-level conclusion. The isolated endpoint result and four prediction increments sit inside distributions dominated by smaller or negative effects, while actual ME–Likert contrast-sign disagreements remain rare.

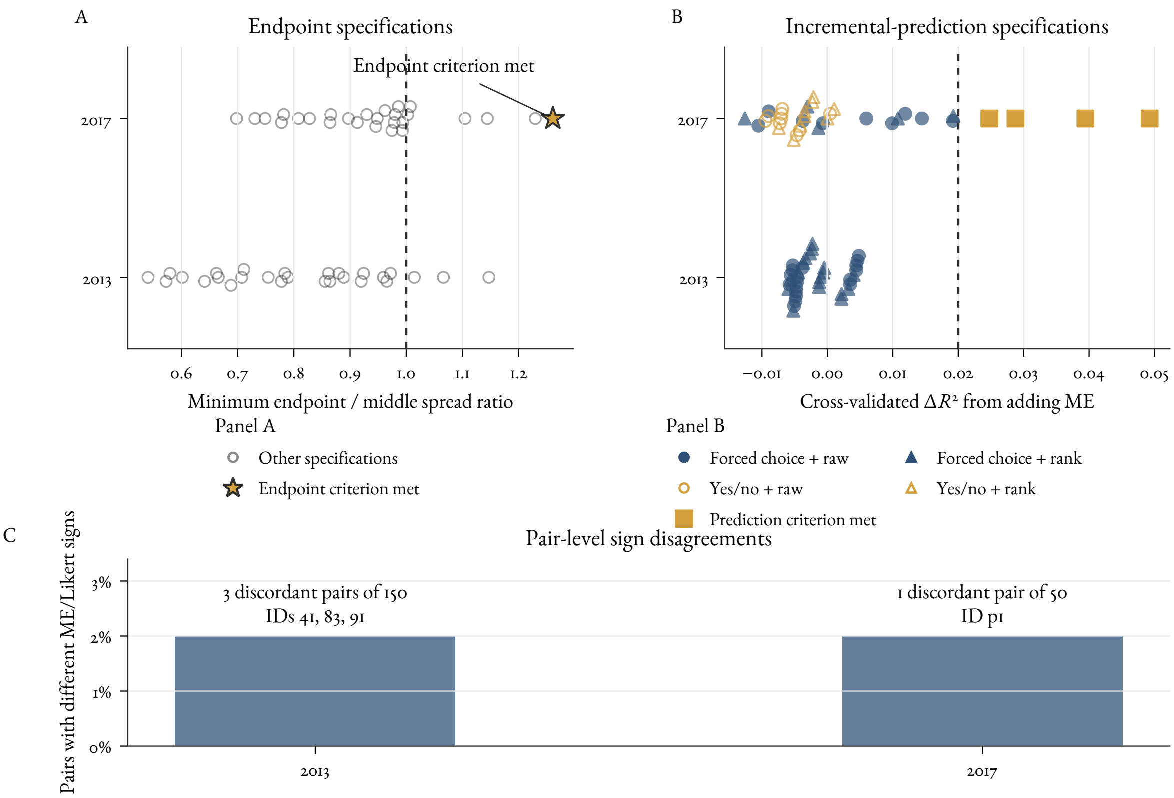

Figure 2: The explicitly post-outcome multiverse. Panel A shows all 54 admissible endpoint specifications; the dashed line marks equal minimum endpoint and middle spread. The golden star is the sole specification in which both endpoint ratios reach 1.25 and the permutation probability is below .05. Panel B shows all 72 incremental-prediction specifications. Filled blue markers denote a forced-choice target and open golden markers a yes/no target; circles denote raw-scale mappings and triangles rank-scale mappings. Solid golden squares mark the four specifications meeting the prediction criterion: ΔR^2 from adding ME is at least .02 and exceeds the corresponding increment from adding Likert. The dashed line marks the .02 threshold. Symmetric vertical stacking separates nearby specifications; vertical position within a study row has no substantive meaning. Panel C shows the proportion of pairs on which ME and Likert disagree about the contrast sign; it uses pair rows rather than the specification counts in Panels A and B. The disagreements are invariant across all nine ME–Likert scoring combinations: 2013 pair IDs 41, 83, and 91 account for 3 of 150 pairs, and 2017 pair ID p1 accounts for 1 of 50. No decision specification clears its family rule.

These aggregate practical diagnostics are weaker than posterior certainty statements or formal equivalence tests. They show that the available contrasts lack the predicted ME-only pattern; they leave open participant-level method factors, response-style dimensions, and locally dependent item effects.

Type S and Type M errors enter only as properties of estimating a named contrast at a given sample size, not as properties a format carries. The relevant question is whether choosing magnitude estimation rather than another method changes the sign-error probability or the exaggeration ratio for a specific contrast under a design analysis (Gelman & Carlin, 2014). Sensitivity checks are reported as named specifications rather than invisible post hoc choices (Steegen et al., 2016). The multiverse follows that rule, but its timing still makes it a robustness analysis rather than a preregistration.

5 The contemporary afterlife of acceptability

Contemporary benchmark resources show what survived from the formal acceptability programme, but it's worth being exact about what survived. CoLA turns examples from linguistics publications into binary labels for model evaluation (Warstadt et al., 2019); BLiMP turns expert-designed grammatical contrasts into large sets of minimal pairs (Warstadt et al., 2020). What they inherit is the portability of contrasts, the idea that acceptability-related distinctions can be made portable, scored, and compared across systems. They inherit it through the expert-asterisk channel, not through Bard et al.'s naive-participant graded measurement.

They also change the evidential object. CoLA's labels come from published linguistic examples rather than from a new sample of naive participant responses. BLiMP's contrasts are generated from expert-crafted grammars and then used to test whether a model prefers the acceptable sentence in a pair. Those are useful evaluation resources, but they aren't method-comparison data for magnitude estimation, Likert ratings, forced choice, or yes/no judgments.

Resource	Item source	Label channel	Evaluation target
CoLA	Examples from 23 linguistics publications	Binary expert/literature acceptability–grammaticality label	Sentence classification
BLiMP	Expert-crafted grammars, 67 minimal-pair suites	Grammar-generated acceptable/unacceptable contrast	Model preference within a pair
SAD	1,000 sequences from textbooks and Linguistic Inquiry	Separate literature grammaticality and crowd acceptability labels	Channel comparison and acceptability prediction

Table 3: Benchmark construction differs by source, label channel, and evaluation target. These resources show how acceptability-related labels became portable, but they don't compare human elicitation methods.

The Syntactic Acceptability Dataset is the most useful bridge case because it keeps grammatical status and crowd acceptability status apart (Juzek, 2024). That separation is exactly the level discipline this paper needs. Table 3 keeps the split visible: the portability of contrasts survived into CoLA and BLiMP through the expert channel, while Bard's graded naive-participant measurement has a closer analogue in crowdsourced resources like the Syntactic Acceptability Dataset. Benchmark labels show the afterlife of one channel; participant-response datasets are still needed to compare human elicitation methods.

6 Grammaticality and acceptability

The reassessment also needs a disciplined vocabulary. The background view adopted here is that grammar is a system of projectibility profiles (Goodman, 1955): relatively stable patterns that license explicitly scoped inferences from forms, meanings, contexts, speakers, and judgments to uses a community or register treats as available. Research purposes select which inference is worth testing; they don't make the inference true.

The empirical projective claim in this paper is deliberately narrow. For the sampled contrasts in the 2013 and 2017 comparison data, observing relative ordering under one formal elicitation format warrants expecting similar aggregate ordering under another, within the variation reported here. A grammaticality claim has wider targets: interpretation, production, correction, expert classification, register fit, cross-speaker stability, and related consequences. Those targets require data beyond the task-format comparisons analysed in this paper.

The paper distinguishes three observation channels from the target. Participant responses under an acceptability prompt are observations of judgments under a task. Expert asterisks and published example labels are classifications made within a theoretical or editorial practice. Benchmark labels are dataset-construction decisions. The projectibility profile is the target: the structure of inferences a form licenses, what its acceptability, production, correction, expert classification, register fit, and cross-speaker stability jointly project to. Two channels are evidence about the same profile when they bear on the same form's inferential relations.

So the profile isn't the average of the channels, and channel disagreement is not noise to smooth away. When a sentence is rated low by naive participants, left unstarred in the literature, and attested in a register corpus, the verdict about its profile is fixed by which inferences actually hold, tested against further data, not by a vote across channels.

A channel that consistently fails to predict the others establishes divergence, not its interpretation. Independent evidence has to decide whether the divergence reflects task specificity, noise, or a distinct part of the profile. That requirement makes the profile more than an unweighted set of labels and supplies the identity condition the level discipline needs.

This discipline keeps neighbouring constructs apart. Acceptability is one evidential route into a projectibility profile, not the profile itself. Personal production, naturalness, standardness, correction behaviour, and theoretical classification project differently. A sentence can be dispreferred without being ungrammatical, correctable without being unnatural, or standard in one register while unacceptable under another task framing.

The framing earns its place in the measurement argument by predicting where a single task-conditioned acceptability scalar could be insufficient: items where participants rate a sentence low but the literature leaves it unstarred, or where the form is standard in one register and degraded under another task framing. A dataset that samples those channels jointly should put a single-scalar model at risk.

The present comparison matrices contain formal task responses rather than that cross-channel evidence. They test the narrower question of whether ME carries a stable method-specific dimension or practical ordering that the bounded tasks miss. The projectibility account supplies a wider empirical programme, but the paper doesn't treat the task-format result as having already tested it.

Bard et al. helped make one part of this space measurable: graded acceptability judgments from participants (Bard et al., 1996). Their work made a major contribution to the measurement of one evidential channel. It doesn't by itself show that an acceptability scale captures every inference licensed by a grammar. Later benchmarks sometimes repackage acceptability-related labels as grammaticality targets (Warstadt et al., 2019, 2020). The Syntactic Acceptability Dataset is valuable here because it explicitly separates literature-derived grammatical status from crowd acceptability status (Juzek, 2024).

That boundary also limits the no-new-data paper. Existing datasets can compare the stability of elicitation methods and clarify how acceptability labels travel. They can't settle a full projectibility profile for a construction: production, correction, register, interpretation, and community licensing would require data designed for those inferences.

7 Conclusion

A balanced update is straightforward. Bard et al. were right to insist that acceptability could be treated as measurable and methodologically serious. Their documented case for magnitude estimation concerned scale resolution, fine distinctions, naive-participant use, and replication across groups (Bard et al., 1996, p. 32). That claim deserves credit as part of the history of experimental syntax.

Method-specific conclusions require more restraint and a fair test. Magnitude estimation is compared against a measurement null in which later formal tasks read a common item-level acceptability signal for the analysed contrasts through method-specific response functions, and in which the single-dimension assumption is checked before any method-specific term is read. The question is then whether magnitude estimation earns its keep on resolution, the distinctions Bard argued fixed scales censor, or on prediction, decision stability, or construct interpretation, rather than whether it survives a null that books resolution as noise.

The approximation uses aggregate matrices from the two comparison studies, response-function diagnostics, pair-level robustness checks, and a bounded post-outcome multiverse. Its scope matters because the participant rows needed for a full dimensionality model across Bard, Sprouse, and Langsford aren't available to this project.

The reanalysis supports that update. Across the two comparison datasets, magnitude estimation and the bounded methods share a strong aggregate practical signal. First-pass aggregate dimensionality checks show no obvious evidence against a single dominant dimension at that level of aggregation. Raw Likert means show bounded response-function curvature, which is exactly why the null shouldn't treat bounded tasks as simple noisy copies of magnitude estimation.

By contrast, the remaining resolution diagnostics are less favourable to the stronger method claim: endpoint regions don't show unusually large magnitude-estimation spread, and pair-level bounded predictors recover most magnitude-estimation contrast variance. None of the three multiverse families clears its frozen cross-specification criterion in either dataset. Four 2017 rank-scale forced-choice specifications show a small ME increment, but that result doesn't generalize to yes/no judgments, raw-scale mappings, or the 2013 data. Langsford et al. remain article-level evidence because their raw responses aren't available to this project.

Projectibility adds a correspondingly modest payoff. Within the sampled contrasts in the two comparison datasets, observing relative ordering under one formal task warrants expecting similar aggregate ordering under another. Magnitude estimation alone doesn't support an additional robust practical ordering for those contrasts. The result warrants a task-format inference, not the claim that every grammatical inference travels across speakers, registers, or observation channels.

The conclusion would change under a clear ME-only pattern: endpoint regions with much larger ME spread than middle regions, high-ME contrasts that bounded methods consistently placed low, stable method-specific residual structure across scoring choices, or lower sign-error and exaggeration risk for named linguistic contrasts. Those are the patterns the present diagnostics do not show.

Across these checks, the central conclusion remains stable. The durable legacy is the formal measurement of graded acceptability as one route into grammar's projectibility profiles. The broader privileged status sometimes read into magnitude estimation is a separate claim, and later evidence has to earn it separately.

Data and code availability

The manuscript source, analysis and figure-generation code, analysis charter, source- and dataset-verification records, and reproduction instructions are available in the public GitHub repository. No raw participant-level data is committed there. Jon Sprouse approved reuse for the limited secondary methodological analysis described in the charter after checking with Diogo Almeida.

The manuscript reports summary diagnostics rather than redistributing participant-level rows or derived participant-level tables. Scripts generate ignored local structural derivatives, readiness manifests, and descriptive item/contrast signal, dimensionality, response-function, and pair-level robustness outputs under `data/derived/`. The post-outcome multiverse adds aggregate unit and pair scores, specification-level results, and input hashes there; participant-level raw files and rows remain outside the repository.

The local reproduction recipe is documented in `analysis/README.md`. The analysis charter was versioned in Git and fixed before outcome analysis as an internal analysis record rather than an external registry entry.

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